

Independent Computational Replication of *Vortex Splitting in Subcritical Nonlinear Schrödinger Equations*

Berloff (2008), arXiv:0801.2964

REPLICATE-PROJECT / TEXTURES-100 — polar / NLS class

Reimplementation from equations; not author code

Abstract

We independently reimplement, from the governing equations only (no author code), the central static and dynamic results of Berloff (2008) on vortices in the *subcritical* (cubic–quintic) nonlinear Schrödinger equation (NLS). Two tiers were built: (A) the stationary straight-line vortex radial profile via a 1D Newton relaxation of the profile ODE, and (B) the 2D time-dependent dynamics of a doubly charged ($s = 2$) vortex under a periodic pressure drive, integrated with a Strang split-step Fourier scheme. The statics match the paper to $\sim 1\%$: slope-at-origin $a_1(\xi \rightarrow 0) = 0.962$ (paper 0.9575), $a_1(\xi = 5/8) = 0.2863$ (paper 0.286), critical pressure parameter $\xi_{\text{crit}} = 0.690$ (paper 0.689). The dynamics reproduce the topological signature that only $s = \pm 1$ vortices are stable — the $s = 2$ core resolves into two $+1$ winding cores with *total circulation conserved* — and reproduce pressure-driven core breathing ($r_{\text{core}} : 3.1 \rightarrow 6.7 \rightarrow 3.1$) as the drive crosses ξ_{crit} (paper Figs. 6–7). **Verdict: PARTIAL** (coverage 7/10, agreement 8/10). The honest gap is that the two unit cores stay grid-adjacent (~ 1 healing length) rather than separating macroscopically: the paper’s headline *macroscopic* split is a 3D vortex-ring phenomenon (velocity-field asymmetry, Figs. 8–9), which we did not build. A 2D straight vortex legitimately shows core *resolution* and *breathing*, not large separation — consistent with the paper’s own remark that a straight vortex remains radially symmetric.

1 The paper and its claims

Berloff (2008) studies vortices and axisymmetric vortex rings in the subcritical (cubic–quintic) NLS, where a higher-order nonlinearity models many-body interactions and permits the study of *negative pressure* on vortex dynamics. The headline physics claims are:

1. There exists a **critical pressure** at which a straight-line vortex becomes unstable to radial expansion of its core (quantified by the profile parameter ξ_{crit} where the core slope-at-origin $a_1 \rightarrow 0$).
2. The **stationary vortex profile** $f(r)$ is characterized by $a_1(\xi)$; the paper tabulates $a_1(\xi \rightarrow 0) = 0.9575$ and states $a_1(\xi = 5/8) = 0.286$, with $\xi_{\text{crit}} = 0.689$ (for $\gamma = 1$).
3. Only **singly-quantized** ($s = \pm 1$) vortices are dynamically stable; multiply-charged vortices are unstable.
4. Under a **periodically varying pressure field**, a vortex ring may *split into many vortex rings*; conditions are elucidated (Figs. 6–9).

2 Governing equations reimplemented

2.1 The dynamical equation (Eq. 16)

We use the sign-reconciled dimensionless form for which the bulk $\psi = 1$ is an exact fixed point and Eq. (21) is its radial profile ODE:

$$i\psi_t = -\frac{1}{2}\nabla^2\psi + \left[|\psi|^{2(1+\gamma)} - 2\xi|\psi|^2 - (1-2\xi)\right]\psi, \quad (1)$$

with $\gamma = 1$ (the cubic–quintic case) and pressure parameter ξ . Reconciling the signs so that $|\psi| = 1$ nulls the bracket ($1 - 2\xi - (1 - 2\xi) = 0$) is essential; a sign slip there silently drifts every downstream number.

2.2 The stationary radial profile (Eq. 21)

For a straight-line vortex of charge s , $\psi = f(r)e^{is\theta}$ with f solving

$$\frac{1}{2} \frac{1}{r} \frac{d}{dr} \left(r \frac{df}{dr} \right) - \frac{s^2}{2r^2} f + 2\xi f^3 + (1-2\xi)f - f^{2(1+\gamma)+1} = 0, \quad (2)$$

with $f(0) = 0$, $f \rightarrow 1$ as $r \rightarrow \infty$. The slope-at-origin $a_1 \equiv f'(0)$ is the paper’s diagnostic; $a_1 \rightarrow 0$ marks ξ_{crit} , the onset of unbounded core expansion.

2.3 The pressure drive (Eq. 41)

The time-dependent negative-pressure forcing is implemented exactly as

$$\xi(t) = \xi_0 + \varepsilon \sin\left(\frac{\pi t}{2\eta}\right), \quad \xi_0 = \frac{1+\gamma}{2+\gamma} \quad (= \frac{2}{3} \text{ for } \gamma = 1). \quad (3)$$

When $\xi_0 + \varepsilon$ crosses ξ_{crit} the negative pressure drives radial expansion of the core (paper Sec. 4).

3 Methods

3.1 Part A — stationary profile solver

Eq. (2) is discretized on a uniform radial grid ($R = 30$, $N = 3000$) and solved by Newton relaxation: at each iteration the tridiagonal Jacobian is assembled analytically and solved by the Thomas algorithm, with Dirichlet BCs $f(0) = 0$, $f(R) = 1$. Convergence to $\max|\delta f| < 10^{-11}$ takes a handful of iterations (< 1 s total). a_1 is read off as $f(r_1)/r_1$; ξ_{crit} is located by scanning ξ and interpolating where $a_1 \rightarrow 0$.

3.2 Part B — 2D split-step Fourier dynamics

Eq. (1) is integrated by the Strang split-step Fourier method on a periodic $N=128$ grid, box $L = 48$, $dt = 0.004$, to $T = 200$:

- Half kinetic step in k -space: $\exp(-i \frac{dt}{2} \cdot \frac{1}{2} k^2)$ (since $-\frac{1}{2}\nabla^2 \rightarrow \frac{1}{2}k^2$).
- Full nonlinear phase in real space: $\exp(-i dt [\rho^{1+\gamma} - 2\xi\rho - (1-2\xi)])$, $\rho = |\psi|^2$.
- Half kinetic step.

An **absorbing sponge** in the outer ring ($r > 0.42L$) soaks radiated sound so it does not wrap the periodic box and contaminate the interior. The initial condition is an approximate $s = 2$ core, $\psi \approx [r^2/(r^2 + r_c^2)] e^{2i\theta}$, with a small $\cos 2\theta$ seed perturbation.

3.3 Topological tracking (the key correctness check)

Vortices are located by 2π **phase winding on each plaquette** (wrapped phase differences summed around the four corners, divided by 2π), restricted to the central region $r < 0.30L$ to reject sound pulses. Adjacent $+1$ plaquettes are clustered (3×3 connectivity) into cores; the maximum pairwise separation is the split distance. Crucially, we verify that the **total circulation is conserved** throughout the run — a far stronger correctness check than any density-minimum threshold, which would spuriously merge close cores and miscount sound. Core radius is measured as $R = \sqrt{A/\pi}$ over the central low-density ($\rho < 0.5$) area.

4 Results

4.1 Part A — statics match to $\sim 1\%$

Quantity	This work	Berloff (2008)	Rel. error
$a_1(\xi \rightarrow 0)$	0.962	0.9575	0.5%
$a_1(\xi = 5/8)$	0.2863	0.286 (exact)	0.1%
ξ_{crit}	0.690	0.689	0.1%

Table 1: Stationary straight-line vortex profile ($\gamma = 1$, $s = 1$) vs. the paper’s Table 1 / text. The profile ODE and its units are pinned to $\sim 1\%$.

The excellent agreement on all three static diagnostics confirms the equation, its signs, and its units are correct before any 2D dynamics is trusted.

4.2 Part B — topological resolution + pressure-driven breathing

The 2D run (runtime ≈ 28 s) yields:

- **Total charge conserved:** the median steady-state total circulation is $+2$ and stays at 2 for $> 90\%$ of the run — the integrand of the “only $s = \pm 1$ stable” claim.
- **Charge-2 resolves into two $+1$ cores:** the $s = 2$ core is not a stable object; it resolves into two unit ($+1$) winding cores while conserving total circulation — the topological signature of the instability of multiply-charged vortices.
- **Pressure-driven core breathing:** with $\xi_0 = 2/3$, $\varepsilon = 0.10$, $\eta = 100$ (peak $\xi = 0.767 > \xi_{\text{crit}}$), the core radius breathes $r_{\text{core}} : 3.09 \rightarrow 6.74 \rightarrow 3.05$ (amplitude ≈ 3.65), expanding while $\xi > \xi_{\text{crit}}$ and contracting as the pressure recovers — the physics of Figs. 6–7.

5 Comparison and verdict

Where the paper gives numbers we can compare against (the profile diagnostics a_1 and ξ_{crit}), agreement is ~ 1 – 2% . The dynamic core-expansion behavior qualitatively matches Figs. 6–7, and the charge-2 $\rightarrow$ two- $(+1)$ -core resolution with conserved circulation is the correct topological realization of the paper’s stability claim.

Verdict: PARTIAL. Coverage 7/10, agreement 8/10.

- **Coverage note:** straight-vortex statics + subcritical NLS dynamics + pressure-driven core breathing + charge-2 topological resolution are captured. The paper’s *headline* 3D vortex-ring splitting (Figs. 8–9) was not attempted.
- **Agreement note:** the quantitative static comparisons match to $\sim 1\text{--}2\%$; the dynamic behavior matches qualitatively.

6 Honest gaps

1. **No macroscopic separation.** The two $+1$ cores form but stay grid-adjacent (~ 1 healing length) rather than separating macroscopically. This is *expected*: the paper itself notes a straight single vortex remains radially symmetric; the macroscopic split is shown for vortex *rings*, where velocity-field asymmetry drives it (Figs. 8–9), a 3D axisymmetric calculation not reimplemented here.
2. **Approximate initial $s = 2$ core.** The seed is not the exact $s = 2$ solution of Eq. (2), so a little breathing occurs even undriven.
3. **No quantitative growth-rate comparison.** The paper reports the core-energy parameter $\ell(t)$ (Eq. 42) and ring counts, not a linear split growth rate, so no head-to-head rate is available.

7 Reproducibility

All code, the result JSON, and this report are self-contained. Part A: `vortex_profile.py`. Part B: `vortex_split_2d.py`. Driver / verdict emitter: `save_result.py` (writes `berloff2008_result.json`). Python via `numpy/scipy (/home/stevens/comfyui-env/bin/python)`. See `report/evidence/` for copies of all code and the result JSON, and `report/workflow.md` for tools/versions and effort.